

NetPath

High-Level Functional and Non-Functional Requirements

Standalone Requirements Document | Based on Proof of Concept | May 2026

1. Purpose

This document defines high-level functional and non-functional requirements for NetPath, a web-based interactive network education platform. It is intended to guide early product planning, system analysis, architecture decisions, and future detailed requirements elaboration.

2. Project Scope Summary

NetPath teaches networking fundamentals through guided lessons, a drag-and-drop topology sandbox, real-time validation, and animated request simulation. The platform supports individual learners, instructors, corporate training teams, academic programs, and platform administrators. Initial curriculum scope includes LAN fundamentals and WAN/Internet concepts, with later growth into advanced networking modules and institutional integrations.

3. Assumptions

- The initial product is browser-native and requires no desktop simulator installation.
- The simulation emphasizes conceptual topology, device roles, and connectivity rather than low-level packet timing or performance modeling.
- The product follows a freemium model with paid Pro, Teams, and Academic tiers.
- Registered users can persist progress and saved topologies; anonymous users may complete early tutorial content where allowed.
- Institutional and team features are expected to include cohort management, progress reporting, and LMS integration.

4. High-Level Functional Requirements

Functional requirements describe the capabilities the system must provide to users and administrators.

ID	Requirement	Description	Primary Actor(s)
FR-01	Guided Curriculum Delivery	The system shall present a progressive networking curriculum organized into LAN and WAN arcs, with short stages that follow an explain, demonstrate, practice, validate loop.	Learner
FR-02	Stage Unlocking	The system shall unlock subsequent stages only after the learner successfully completes the required interactive task for the current stage.	Learner

ID	Requirement	Description	Primary Actor(s)
FR-03	Device Palette	The system shall provide a palette of unlocked networking devices, including hosts, hubs, switches, routers, access points, DNS servers, firewalls, and NAT gateways as appropriate to the learner's current stage.	Learner
FR-04	Drag-and-Drop Topology Builder	The system shall allow learners to drag devices onto a canvas and arrange them into network topologies.	Learner
FR-05	Connection Drawing	The system shall allow learners to create wired links between device ports and wireless associations through a dedicated interaction mode.	Learner
FR-06	Device Configuration	The system shall provide a lightweight configuration panel for lesson-appropriate settings such as IP address, default gateway, and DHCP toggle.	Learner
FR-07	Real-Time Validation	The system shall continuously validate the learner's topology against stage success criteria and provide color-coded status feedback for valid, broken, or partially complete connectivity.	Learner
FR-08	Request Simulation	The system shall allow learners to send a simulated request through a validated topology and view the request and response path device by device.	Learner
FR-09	Simulation Explanation Tooltips	The system shall display explanatory tooltips at each step of a simulated request path describing the forwarding or routing decision at that device.	Learner
FR-10	Undo, Redo, and Reset	The system shall support undo, redo, and full reset actions in the sandbox so learners can experiment safely.	Learner

ID	Requirement	Description	Primary Actor(s)
FR-11	Progress Persistence	The system shall save registered learners' stage progress, earned badges, unlocked devices, and completed challenge status.	Learner
FR-12	Topology Portfolio	The system shall allow learners to save and revisit completed or custom sandbox designs in a personal topology portfolio.	Learner
FR-13	Gamification and Badges	The system shall award illustrated device cards or stage badges when learners complete designated curriculum milestones.	Learner
FR-14	Timed Challenge Mode	The system shall provide optional timed challenge modes for advanced learners or certification preparation.	Learner
FR-15	Account Registration and Authentication	The system shall support user registration, login, session management, and account-based progress recovery.	Learner, Admin
FR-16	Freemium Access Control	The system shall enforce tier-based access to free, pro, team, and academic features, including limits on arcs, stages, saved designs, and advanced challenges.	System Admin
FR-17	Cohort and Team Management	The system shall allow team or academic administrators to create cohorts, invite users, and view learner membership.	Instructor, Team Admin
FR-18	Progress Tracking Dashboard	The system shall provide instructors or team administrators with aggregate and individual learner progress views.	Instructor, Team Admin
FR-19	Custom Challenge Creation	The system shall allow authorized team or academic users to create or assign custom sandbox challenges.	Instructor, Team Admin
FR-20	LMS Integration	The system shall support integration with learning management systems through an LTI-compatible connector for institutional adoption.	Institutional Admin

ID	Requirement	Description	Primary Actor(s)
FR-21	Shareable Network Outputs	The system shall generate shareable visual cards or links for completed sandbox designs when sharing is enabled.	Learner
FR-22	Curriculum Administration	The system shall allow authorized platform staff to create, update, publish, and retire curriculum stages, device definitions, and validation rules.	Platform Admin
FR-23	Analytics and Reporting	The system shall collect usage, activation, completion, retention, and conversion metrics needed to monitor the funnel and product performance.	Platform Admin
FR-24	Email Engagement	The system shall support lifecycle email nudges such as progress reminders and challenge notifications for registered users who opt in.	Learner, Platform Admin

5. High-Level Non-Functional Requirements

Non-functional requirements describe measurable qualities and constraints. Each requirement below includes objective verification or validation criteria.

ID	Quality Attribute	Requirement	Objective Verification / Validation Criteria
NFR-01	Usability	The platform shall be usable by beginning networking learners without prior simulator experience.	In moderated usability testing with at least 8 target learners, 80% shall complete Stage 1 without facilitator assistance within 10 minutes.
NFR-02	Performance - Validation Feedback	Topology validation feedback shall feel immediate during canvas editing.	For standard lesson topologies with up to 50 nodes and 75 links, validation status shall update within 250 ms after a user edit in 95% of test runs.
NFR-03	Performance - Simulation Playback	Request simulation animations shall play smoothly on supported devices.	During a 20-hop simulation on a supported desktop browser, animation shall maintain at least 45 frames per second for 95% of playback duration.
NFR-04	Availability	The hosted production service shall be available for learners and administrators during normal operating periods.	Monthly uptime for the web application and API shall be at least 99.5%, excluding scheduled maintenance announced at least 24 hours in advance.
NFR-05	Scalability	The system shall support growth from beta to public launch usage without redesigning core architecture.	Load testing shall demonstrate support for 1,000 concurrent active users with 95th percentile API response time at or below 500 ms for common authenticated requests.
NFR-06	Browser Compatibility	The platform shall run in modern browsers without local installation.	Core learner flows shall pass regression tests on the latest stable versions of Chrome, Edge, Firefox, and Safari on desktop; no blocking defects allowed.
NFR-07	Responsive Layout	The learner experience shall support desktop and tablet form factors.	All Stage 1-10 learner flows shall be usable at viewport widths of 1024 px and 1366 px with no horizontal scrolling except inside the sandbox canvas.

ID	Quality Attribute	Requirement	Objective Verification / Validation Criteria
NFR-08	Accessibility - Keyboard	The platform shall support keyboard navigation for primary learning and sandbox interactions.	A tester shall be able to complete Stage 1 using only a keyboard, including navigating lesson content, adding devices, connecting them, validating, and sending a request.
NFR-09	Accessibility - Screen Reader Labels	The canvas and interactive network elements shall expose meaningful labels to assistive technologies.	Automated accessibility checks shall report no critical ARIA labeling violations on core lesson and sandbox pages; manual screen-reader review shall confirm device and link labels are announced.
NFR-10	Accessibility - Visual Design	The product shall support colorblind-safe interpretation of validation states.	Validation status shall use both color and a non-color indicator, and UI contrast for text and controls shall meet WCAG 2.1 AA contrast thresholds.
NFR-11	Accessibility - Reduced Motion	The product shall respect reduced-motion preferences.	When the operating system or browser reduced-motion preference is enabled, request-path animation shall be replaced by a non-animated or minimally animated step display.
NFR-12	Security - Authentication	Authenticated user areas shall protect account and progress data from unauthorized access.	Security testing shall verify that unauthenticated requests cannot access protected API endpoints and that one user cannot retrieve or modify another user's progress records.
NFR-13	Security - Transport Protection	All production web traffic shall be encrypted in transit.	Production endpoints shall enforce HTTPS and receive an A or better rating in a standard TLS configuration scan.
NFR-14	Privacy	The platform shall collect only data necessary for learning progress, account management, analytics, and subscribed communications.	A data inventory review shall show each stored personal-data field has a documented purpose, retention category, and owner; no undocumented personal-data fields may be present in production schemas.
NFR-15	Reliability - Data Persistence	Learner progress and saved topologies shall not be lost during normal use.	In regression testing, 100 consecutive save and reload operations for progress and saved topologies shall complete with no data loss or corruption.

ID	Quality Attribute	Requirement	Objective Verification / Validation Criteria
NFR-16	Recoverability	The system shall be recoverable from data-store or deployment failure.	A restoration drill from backup shall recover the database to within 24 hours of the failure point and restore service in a staging environment within 4 hours.
NFR-17	Maintainability	The codebase shall support ongoing curriculum and feature expansion by a small team.	Continuous integration shall run automated unit and integration tests on every merge request, and the main branch shall maintain at least 80% coverage for validation-engine logic.
NFR-18	Testability	Core topology validation rules shall be objectively testable.	Each published lesson validation rule shall include automated positive and negative test cases before the lesson can be released.
NFR-19	Extensibility	The architecture shall allow new devices and lesson stages to be added without rewriting the simulation engine.	A new device type with configuration fields and validation behavior shall be addable through documented extension points with no changes to unrelated device implementations.
NFR-20	Localization Readiness	The user interface and curriculum content shall be structured to support future translation.	A localization audit shall verify that user-facing strings for navigation, lesson text, errors, and buttons are externalized from source code for the core learner flows.
NFR-21	Observability	The system shall provide operational visibility into errors and key product events.	Production shall capture structured logs, error traces, and metrics for API errors, validation failures, stage completions, sign-ups, and payment conversion events.
NFR-22	Compliance - LMS Interoperability	Institutional integration shall align with common LMS launch and roster workflows.	An LTI integration test shall successfully launch a NetPath activity from at least one supported LMS test environment and return learner activity status as configured.

6. Requirement Traceability Notes

The requirements in this document trace to the NetPath proof of concept sections describing the guided tutorial stages, interactive sandbox, request simulation, progression and gamification, accessibility commitments, technology stack, customer acquisition model, teams and academic tiers, LMS integrations, and long-term growth levers.

7. Recommended Next Steps

1. Prioritize requirements into MVP, beta, and post-launch releases.
2. Convert high-priority functional requirements into detailed user stories with acceptance criteria.
3. Create a requirements traceability matrix linking requirements to use cases, design components, tests, and release milestones.
4. Validate non-functional targets with stakeholders before architecture and vendor decisions are finalized.